Points 6 Published

Details Questions

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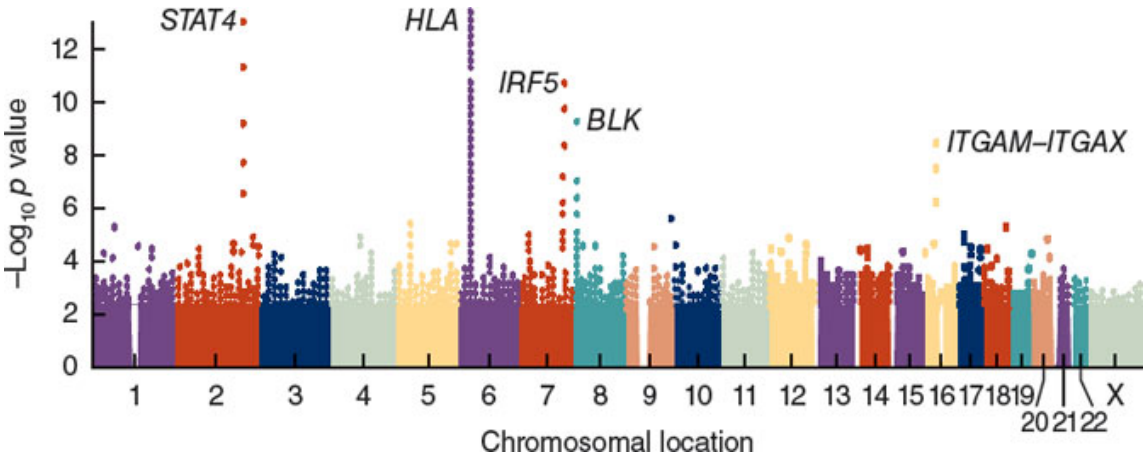
GWAS Questions

Systemic lupus erythematosus (SLE) is a chronic autoimmune disease that can lead to inflammation and tissue damage in any part of the body and even death. A large-scale GWAS analysis of this condition was perfomed; the study involved the genotyping of more than 500,000 SNP loci in more than 1000 Cases (patients) and 3000 Controls of European ancestry. The data revealed significant associations with five genetic loci, whose names and positions are shown in the Manhattan plot that follows.



GWAS Question 0.43 pts

Are any of these five loci genetically linked?

- Yes
- No

GWAS Question 0.43 pts

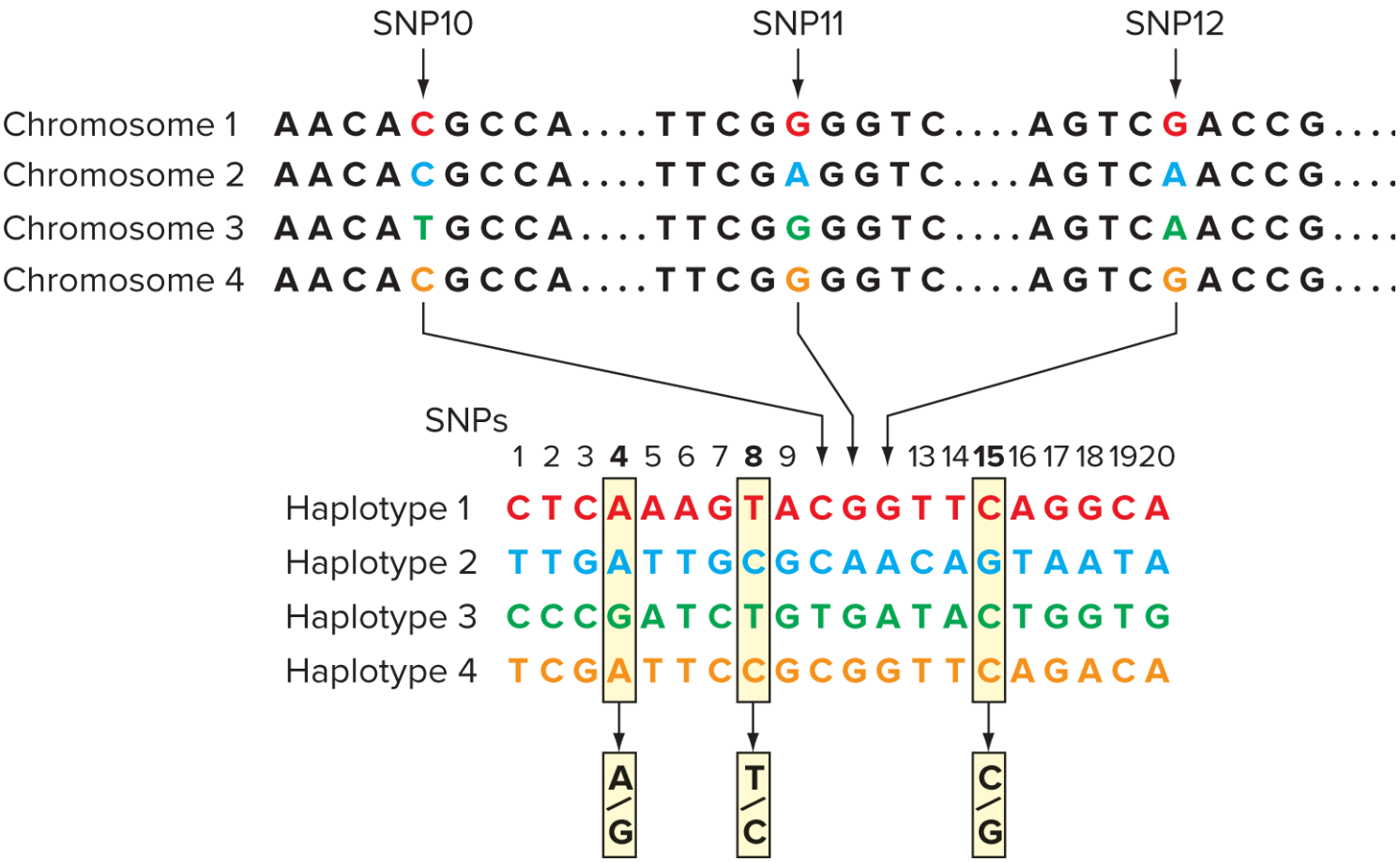
Which p-value threshold would indicate that the difference between the frequency of the allele in the patient population and in the control population is significant?

- 0.05
- 0.01
- 0.001
- 0.0000005
- 0.0000001
- 0.00000005
- 0.00000001

GWAS Question

In GWAS analysis, because of the existence of LD blocks (or haplotype blocks), it is not necessary to genotype a person for every one of the 50 million known common SNPs. Haplotype blocks are stretches of DNA containing particular SNP variants that tend to be inherited together (as a block) because recombination within the region is rare. In the accompanying figure, three different SNP loci are shown at top (SNP10, SNP11, SNP12), each with two alleles among the world's population of humans.

Only four of all the possible combinations of these SNP alleles are found in human genomes, as shown in the four chromosome types pictured. These three SNPs are part of a large block of 20 SNPs that are usually inherited in one of the four configurations, or haplotypes, shown. Because these 20 SNPs are inherited as haplotype blocks, genotyping any individual for the three so-called Tag SNPs (SNP4, SNP8, and SNP15 shown in bold) should be sufficient to predict that individual's alleles for the other 17 SNPs.



GWAS Question 0.43 pts

How many configurations of the three SNPs shown at the top of the diagram (SNPs 10, 11, and 12) are theoretically possible?

8

GWAS Question 0.43 pts

You genotype an individual using the Tag SNPs on a SNP microarray. For SNP4, only the A allele was present. For SNP8, only the C allele was present. For SNP15, on the G allele was present.

What is the haplotype of this individual?

- 1/1
- 2/2
- 3/3

- 4/4

GWAS Question 0.43 pts

You genotype another individual using a SNP microarray. For SNP4, only the A allele was present. For SNP8, both the T and C allele were present. For SNP15, only the C allele was present.

What is the haplotype of this individual?

- 1/1
- 1/2
- 1/3
- 1/4

GWAS Question

You conduct a Case/Control study comparing the frequency of a single SNP (with alleles C and T) among individuals who have developed high blood pressure as compared with control individuals of a similar age who show no sign of this condition. You obtain the following results:

	Cases	Controls
C	1025	725
T	902	922

GWAS Question 0.84 pts

Does a statistically significant association exist between this SNP and the risk of developing high blood pressure, if this is the only SNP that you test?

Yes
No
Borderline significance
Not enough information

GWAS Question 0.43 pts

Would this association still be significant if this was one of a million SNPs that you tested in your study?

- Yes
- No
- Borderline significance
- Not enough information

GWAS Question 0.43 pts

Which allele (C or T) is associated with a higher risk of developing high blood pressure?

- C
- T

GWAS Question 0.43 pts

Calculate the allelic odds ratio for the risk allele. What does the alleleic odds ratio mean about a person's chance of developing high blood pressure? Give your answer rounded to two decimal places.

1.44

1.45

QTL Question 0.43 pts

Two alleles at one locus produce three distinct phenotypes. Two alleles of two genes lead to five distinct phenotypes. Two alleles of six genes lead to 13 distinct phenotypes. (These statements assume that the alleles at any one locus are codominant or incompletely dominant and that each gene makes an equal contribution to the phenotype.) Which of the following is a formula to express the relationship between number of genes and number of phenotypes? Let *n* equal the number of genes.

n

2n

2n + 1

3n

3n + 1

QTL Question 0.43 pts

Each of the most extreme phenotypes for a trait determined by two alleles at one locus are found in a proportion of 1/4 in the F_2 generation. If two alleles of two genes determine the trait, each extreme phenotype will be present in the F_2 as 1/16 of the population.

In common wheat (*Triticum aestivum*) kernel color varies from red to white and the genes controlling the color act additively, that is, alleles for each gene are incompletely dominant and each gene contributes equally to the color. A true-breeding red variety is crossed to a true-breeding white variety, and 1/256 of the F_2 have red kernels and 1/256 have white kernels. How many genes control kernel color in this cross?

4

QTL Question 0.43 pts

Discussion (Chapter 22, Problem #16):

In the QTL study described on page 759 of Hartwell et al., the fruit weight 2.2 locus (fw2.2) was the largest effect QTL - the Solanium lycopersicum (large) allele explained up to 30% of fruit weight variance when compared with the effects of the Solanium pennellii (small) allele.

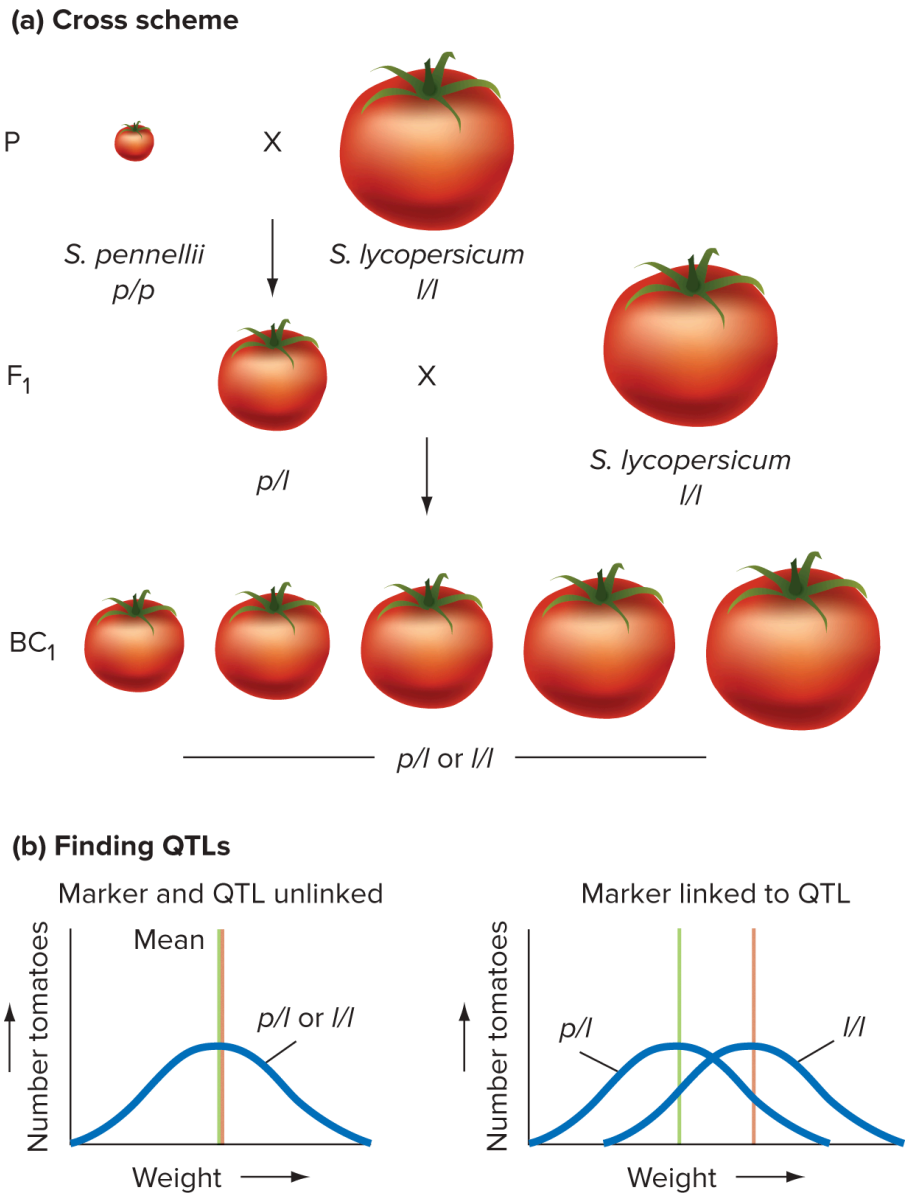
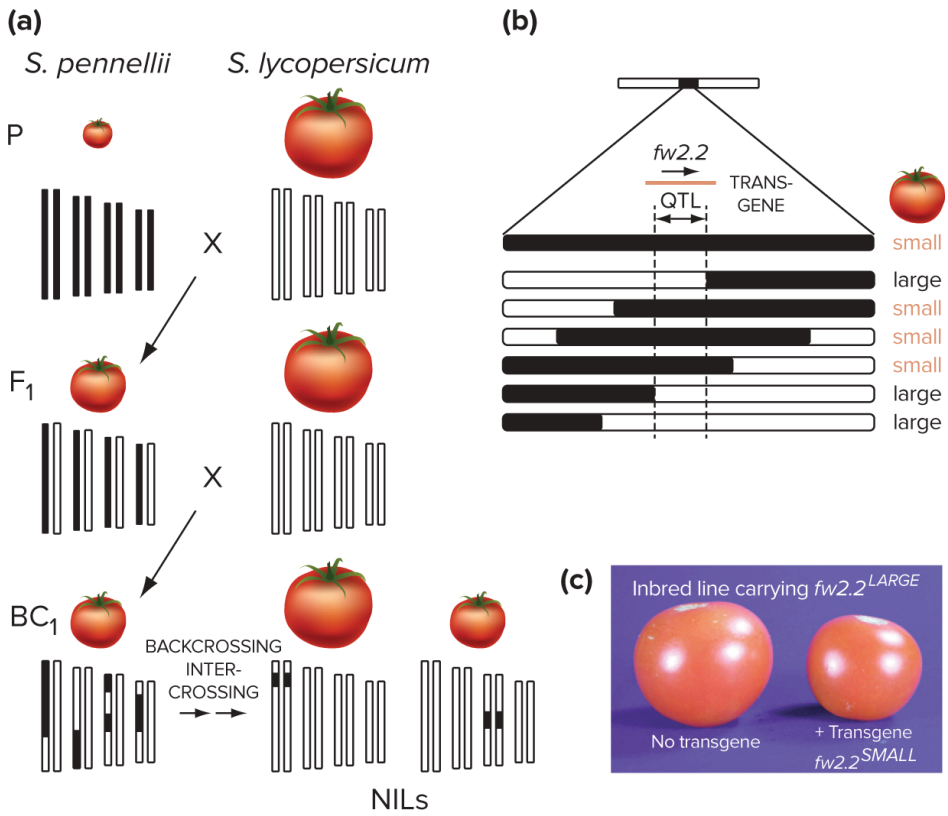


Figure 22.13 Strategy for genetic mapping of quantitative trait loci (QTLs) influencing complex traits. (a) Isogenic *S. pennellii* (*p/p*) and *S. lycopersicum* (*l/l*) parents were crossed. The resulting F_1 generation was then backcrossed to isogenic *S. lycopersicum*, creating the BC_1 generation. **(b)** BC_1 individuals were weighed and genotyped. If a significant difference in trait value (here, the weight) exists between heterozygotes with the *S. pennellii* and *S. lycopersicum* alleles of a marker and homozygotes for the *S. lycopersicum* allele, the marker is linked to a QTL influencing the phenotype.

To validate that the fw2.2 gene was the main causal gene for tomato size, researchers cloned the *S. pennellii* allele of the fw2.2 gene and introduced it as a transgene into *S. lycopersicum*. This is described in Figure 22.13:



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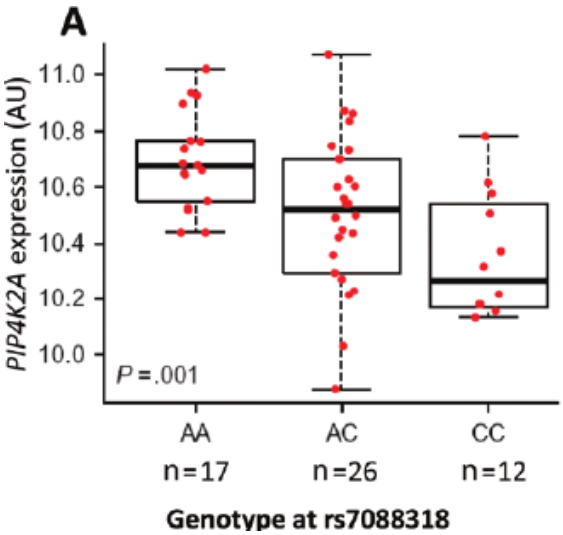
Why was the experiment in A and B done this way (by crossing the S. pennellii allele into the S. lycopersicum background), rather than the other way around? Identify all of the correct answers.

Phenotypic rescue (of ancestral, small fruit size) needs to be tested in the organism of interest, i.e., the one with the novel trait.
No reason. The other way would be as effective.
The design of the QTL screen: p/l heterozygotes were tested for reduced size compared to l/l homozygotes. The experiment did not directly test for QTLs that increase fruit size.
S. pennellii is difficult to grow

QTL Question 0.43 pts

You performed a genome-wide study in lymphoblastoid cell lines derived from 55 individual donors to identify eQTLs. You examined 709509 SNPs. Using linear regression, you find an association of genotypes at rs7088318 with expression of the gene *PIP4K2A*. Shown below is a plot of the association along with the p-value of the association. AU = arbitrary units.

Is this a significant association?



Yes

No

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